L22 — Stack: Basic Terminology, Sequential and Linked Representations

Unit: 2 | **Chapter:** Stacks & Recursion | **BT Level:** BT3 | **CO:** CO2

Learning Objectives

- Define a stack and its LIFO principle
- Implement a stack using arrays (sequential representation)
- Implement a stack using a linked list
- Compare both implementations and describe their trade-offs

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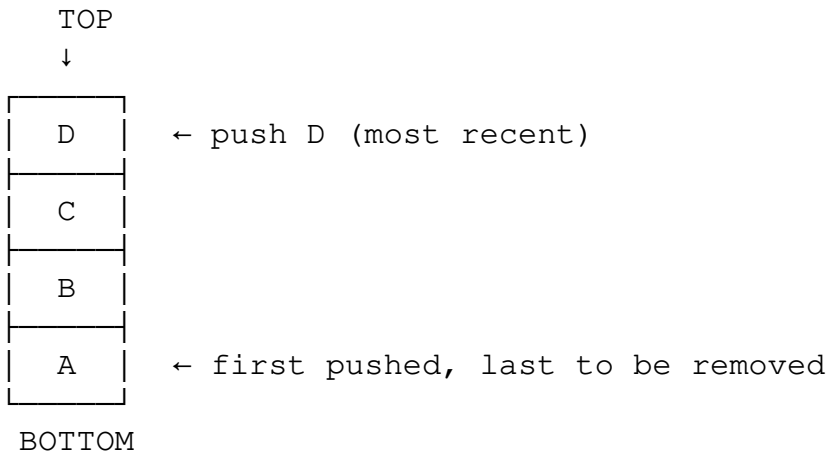
1. What is a Stack?

A **stack** is a linear data structure that follows the **LIFO (Last In, First Out)** principle — the element inserted last is the first to be removed.

Real-world analogy: A stack of plates — you add and remove from the top only.



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2. Stack Terminology

Term	Definition
TOP	Index/pointer to the topmost element
PUSH	Insert element onto the top of the stack
POP	Remove element from the top of the stack
PEEK / TOP	View top element without removing
isEmpty	Check if stack has no elements
isFull	Check if stack is at maximum capacity (array-based only)
Overflow	Attempting PUSH on a full stack
Underflow	Attempting POP on an empty stack

3. Stack ADT

```
ADT Stack:
  Operations:
    push(item)   → void      — Add item to top
    pop()        → item      — Remove and return top item
    peek()       → item      — Return top item (no removal)
    isEmpty()    → bool      — True if stack is empty
    size()       → int       — Number of elements
```

4. Array-Based (Sequential) Stack

4.1 Implementation

```
class ArrayStack:
    def __init__(self, capacity=100):
        self.stack = [None] * capacity
        self.capacity = capacity
        self.top = -1     # -1 means empty

    def push(self, item):
        if self.top == self.capacity - 1:
            raise OverflowError("Stack Overflow")
        self.top += 1
        self.stack[self.top] = item

    def pop(self):
        if self.is_empty():
            raise IndexError("Stack Underflow")
        item = self.stack[self.top]
        self.stack[self.top] = None  # Optional: clear slot
        self.top -= 1
        return item

    def peek(self):
        if self.is_empty():
            raise IndexError("Stack is empty")
        return self.stack[self.top]

    def is_empty(self):
        return self.top == -1

    def is_full(self):
        return self.top == self.capacity - 1

    def size(self):
        return self.top + 1

    def display(self):
```

```
        print("[", end="")
        for i in range(self.top + 1):
            print(self.stack[i], end=", " if i < self.top else "")
        print("] ← TOP")
```

4.2 Trace

```
s = ArrayStack()
s.push(10)   # stack: [10]           top=0
s.push(20)   # stack: [10, 20]       top=1
s.push(30)   # stack: [10, 20, 30]   top=2
s.peek()     # returns 30, top=2 (unchanged)
s.pop()      # returns 30, stack: [10, 20], top=1
s.pop()      # returns 20, stack: [10],     top=0
```

4.3 Complexity

Operation	Time	Space
push	O(1)	O(1)
pop	O(1)	O(1)
peek	O(1)	O(1)
isEmpty	O(1)	O(1)

Total space: O(n) for n elements.

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5. Linked List Stack

Uses a linked list — no fixed capacity.

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class LinkedStack:
    def __init__(self):
        self.top_node = None   # top of stack = head of linked list
        self._size = 0

    def push(self, item):
        new_node = Node(item)
        new_node.next = self.top_node   # New node points to old top
        self.top_node = new_node        # New node becomes top
        self._size += 1

    def pop(self):
        if self.is_empty():
            raise IndexError("Stack Underflow")
        item = self.top_node.data
        self.top_node = self.top_node.next   # Advance top
```

```
        self._size -= 1
        return item

    def peek(self):
        if self.is_empty():
            raise IndexError("Stack is empty")
        return self.top_node.data

    def is_empty(self):
        return self.top_node is None

    def size(self):
        return self._size

    def display(self):
        curr = self.top_node
        print("TOP ", end="")
        while curr:
            print(f"[{curr.data}]", end=" → ")
            curr = curr.next
        print("None")
```

5.1 Trace

```
push(10):  TOP [10] → None
push(20):  TOP [20] → [10] → None
push(30):  TOP [30] → [20] → [10] → None
pop():     returns 30, TOP [20] → [10] → None
```

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6. Array Stack vs Linked Stack

Property	Array Stack	Linked Stack
Capacity	Fixed (pre-allocated)	Dynamic (unlimited)
Memory	Fixed, may waste	Allocates as needed
Memory overhead	None	Pointer per node (+8 bytes)
Cache performance	Excellent	Poor
Overflow possible	Yes	No (memory permitting)
Implementation	Simpler	Slightly complex
push / pop time	O(1)	O(1)

When to use:

- **Array stack:** Known max size, performance-critical, simple implementation
- **Linked stack:** Unknown/variable size, no memory waste

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7. Python's Built-in Stack

Python's `list` can be used directly as a stack:

```
stack = []
stack.append(10)   # push
stack.append(20)
stack.append(30)
print(stack[-1])   # peek → 30
stack.pop()        # pop → 30
```

- `list.append()` → O(1) amortized
- `list.pop()` → O(1)

Python also has `collections.deque` for a more efficient stack (no resizing overhead).

Summary

- A **stack** follows LIFO: last pushed, first popped.
- Core operations: push, pop, peek — all O(1).
- **Array implementation:** O(1) operations, fixed capacity, excellent cache performance.
- **Linked list implementation:** O(1) operations, dynamic capacity, pointer overhead.
- Python's `list` serves as a built-in stack.

References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.1 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.2 (Springer, 2020)